

PHILLIP JIANG ZHOU

CONTACT INFORMATION

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EDUCATION

Stanford University, Palo Alto, California
PhD, Genetics

September 2026 - Current

Amherst College, Amherst, Massachusetts

September 2020 - May 2024

BA, Mathematics and Biology, *magna cum laude*, Cumulative GPA: 3.90/4.00

Thesis topic: Optimal transport theory

Supervisor: Professor Ryan J. Alvarado

Yale-NUS College, Singapore

January 2023 - May 2023

Study Abroad, Mathematical, Computational, and Statistical Sciences and Life Sciences

PUBLICATIONS

- **Zhou P.J.**, Simon M. V., Yu T., Gligorovski V., Mathis N., Zhao J., Vogt F., Phan Q.V., Ryu J., Pan Q., Tyagi A., Ascher D.B., Schwank G., Pinello L., Cassa C.A., Sherwood R.I. *LDLR variant classification through activity-normalized prime editing screening*, *Circulation*, accepted.
- **Zhou P.J.**, Simon M. V., Yu T., Gligorovski V., Mathis N., Zhao J., Vogt F., Phan Q.V., Ryu J., Pan Q., Tyagi A., Ascher D.B., Schwank G., Pinello L., Cassa C.A., Sherwood R.I. *LDLR variant classification through activity-normalized prime editing screening*, <https://www.biorxiv.org/content/10.64898/2025.12.16.694467v1>.
- Doerr, S., **Zhou P.J.**, Ragkousi K., *Origin and development of primary animal epithelia*, *BioEssays*, Volume 46, Issue 2, February 2024 (DOI: 10.1002/bies.202300150.).

PROFESSIONAL EXPERIENCE

Brigham and Women's Hospital, Boston, MA

July 2024 - June 2026

Research Technician II

- Interrogated cholesterol homeostasis gene variants using CRISPR screening in Dr. Richard Sherwood's laboratory. Characterized variants with flow cytometry and computational chemistry
- Utilized cDNA display for novel high-throughput apolipoprotein functional screening

Amherst College, Amherst, MA

September 2020 - May 2024

Research Assistant

- Investigated epithelial formation in *N. vectensis* to understand cellular self-organization in Dr. Katerina Ragkousi's laboratory
- Developed pipeline for automated nuclear quantification in embryos during a special topics course

Brigham and Women's Hospital, Boston, MA

June 2023 - August 2023

Research Assistant

- Used single-cell RNA sequencing and flow cytometry to study distal lung responses to allergens in Dr. Daniel Dwyer's laboratory
- Performed 5 flow cytometry experiments informed by differential expression analysis and used 8 mouse lungs for scRNA-seq data analysis
- Completed a summer immunology class at Harvard, participated in weekly lab meetings

Cold Spring Harbor Laboratory, Cold Spring Harbor, NY

June 2022 - August 2022

Research Assistant

- Developed interpretability methods for understanding how the neural network Akita predicts human genome architecture from DNA sequences in Dr. Peter Koo's laboratory
- Wrote code to determine how neural network predictions change under *in silico* perturbations

Stowers Institute for Medical Research, Kansas City, MO

June 2021 - July 2021

Summer Scholar

- Visualized development using in situ hybridization and immunostaining to investigate the evolution of metamorphosis in Dr. Matthew Gibson's laboratory
- Designed original microsurgical protocols to ablate the *N. vectensis* apical sensory organ; performed microsurgery on 1,288 larvae
- Presented summer poster to Stowers scientists and at Amherst College

TEACHING EXPERIENCE

- Presenter**, Harvard Biological and Biomedical Sciences **January 2026**
- Organized and presented for the CRISPR screening module for Harvard's GENETIC 390QC Bootcamp: Experimental Approaches in Genetics for 6 graduate students
- Mentor**, Sherwood Lab **February 2025 - June 2026**
- Mentored a master's student in biotechnology for 3 months and a high school student (Top 300 in the Regeneron Science Talent Search) for 2 months, an undergraduate, and a rotation student in the Harvard Bioengineering PhD program
- Teacher's Assistant**, Department of Mathematics **September 2022 - May 2024**
- Served as a TA for Linear Algebra (MATH 271) and Probability (MATH 360)
- Teacher's Assistant**, Department of Biology **September 2022 - May 2024**
- Served as a TA for Cell Biology (BIOL 291) and Developmental Biology (BIOL 221)

CAMPUS INVOLVEMENT

- Founder/President**, Biotechnology Club **September 2023 - May 2024**
- Arranged weekly events to raise awareness of careers in systems and computational biology
- President**, Math Club **September 2021 - May 2024**
- Coordinated weekly events and social events with an average attendance of 10-20 participants and gave talks on mathematical biology
- Member**, Association of Amherst Students Senate **September 2023 - May 2024**
- Served on the library and arts committees, where I provided input into student utilization of library resources and organized a campus-wide art competition
 - Conducted interviews with a student representative from every academic discipline on campus to deposit in college archives
- Member**, Biology Steering Committee **September 2021 - May 2022**
- Coordinated panels to familiarize students with the Amherst biology department and its research
 - Maintained and updated website designed to help students learn more about the biology major

PRESENTATIONS

- Zhou P.J.**, Simon M. V, Vogt F., Yu T., Zhao J., Phan Q.V., Sherwood R.I., *Prime editing screening of LDLR aids disease diagnosis and reveals gain-of-function variants*, 5th Annual Genetics Retreat, Brigham and Women's Hospital Division of Genetics, Waltham, MA, September 26 2025 (Poster).
- Postnikova A., Lee G., Abdelghany A., Alvarez M., **Zhou P.J.**, Katerina R., *Discovery of cell surface proteins essential for epithelial organization in Nematostella vectensis*, 2024 SDB Northeast Regional Meeting, Marine Biological Laboratory, Woods Hole, MA, April 27 2024 (Poster).
- Zhou P.J.**, Case N.A., Pahima, H.T., Dwyer D.F., *Lung epithelial cells dynamically participate in the immune response to allergens*, Harvard Immunology Undergraduate Summer Program, Boston, MA, August 10 2023 (Poster).
- Zhou P.J.**, Toneyan S., Tang A., Kaczmarzyk J., McCandlish D., Koo P., *Seeing 3D genome architecture through the lens of deep learning using interpretability methods*, Cold Spring Harbor Laboratory Undergraduate Research Program Final Presentation, Cold Spring Harbor, NY, August 5 2022 (Presentation).
- Zhou P.J.**, Dewey J., Sabin K., Gibson M., *Dissecting the function and development of the Nematostella apical organ*, Stowers Summer Scholars Poster Session, Kansas City, MO, July 30 2021 (Poster).

HONORS AND AWARDS

- Harvey Blodgett Scholarship**, Amherst College, 2024
- The Harvey Blodgett Scholarship provides financial support to aid student coursework in biology and geology. I used this \$4,500 scholarship to attend the Bumblekite machine learning school in healthcare and biosciences at ETH Zürich.
- Kauffman Fellowship**, Amherst College, 2023
- The goal of the Kauffman Fellowship is to allow a highly self-motivated student to perform biomedical laboratory research in a field of interest, regardless of the laboratory's ability to offer a stipend or the student's financial need.

Walker Award for Leadership, Amherst College, 2022

- The Walker Award for Leadership is awarded for outstanding leadership and for contributions to the mathematics and statistics community at Amherst. Nominations for the Walker Award can be made by students, staff, and faculty.